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Final Report – DSE6111 Predictive Modeling

Business Problem Statement

ABC Corporation is facing customer attrition, which immediately puts its revenue stream, customer lifetime value, and long-term brand loyalty at risk. In a competitive service business, knowing which customers are most likely to churn is essential to guiding effective and timely retention action.
The goal of this project is to design a predictive model to predict customer churn probability. This 0-1 probabilistic measure will enable the company to prioritize customers based on the likelihood of defecting. Risky customers can be addressed with retention strategies like special promotions, early customer assistance, or personalized marketing contact.

Target Variable:

- Attrition_Flag: Binary attribute where 1 = churned customer, 0 = active customer

Business Impact:
ABC Corporation is able to determine potential-to-churn customers earlier in the decision-making process through the predictive churn model. It can save churn at a comparatively lower cost, enhance customer satisfaction, enhance customer lifespans, and enhance resource utilization between support and marketing groups.

Dataset Overview & Exploratory Data Analysis (EDA)

ABC Corporation provided a customer-level dataset with 10,127 rows and 21 variables that are both demographic and transactional. The target variable, Attrition_Flag, is binary, indicating whether a customer has churned (1) or remains active (0).

Dataset Composition

- Numerical Variables (15): Examples include Customer_Age, Credit_Limit, Total_Trans_Amt, and Total_Revolving_Bal
- Categorical Variables (6): Examples include Gender, Education_Level, Marital_Status, and Card_Category

Target Class Distribution
The target variable is heavily imbalanced:

- 83.9% of customers remain active
- 16.1% of customers have churned

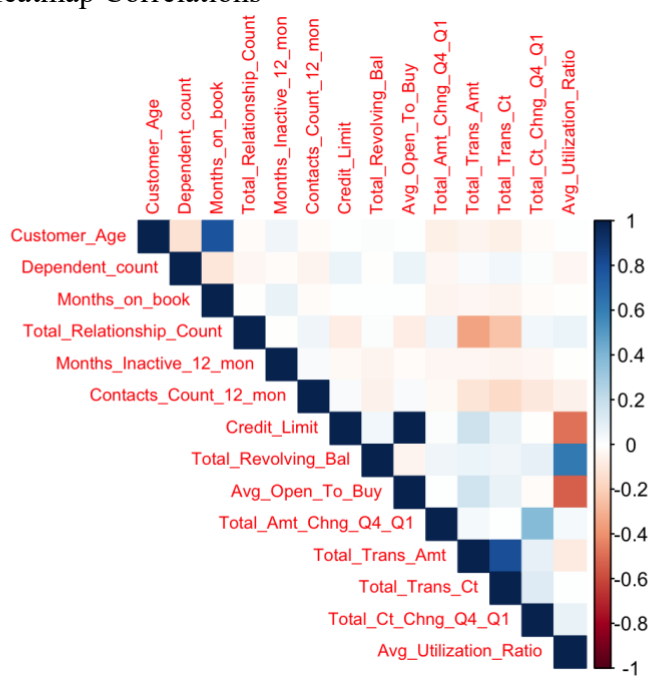
This imbalance was managed in the modeling stage by implementing stratified sampling and class weighting so that the model will never be biased toward the majority class.

Exploratory Analysis Findings
The important correlations were established by heatmaps and correlation matrices:

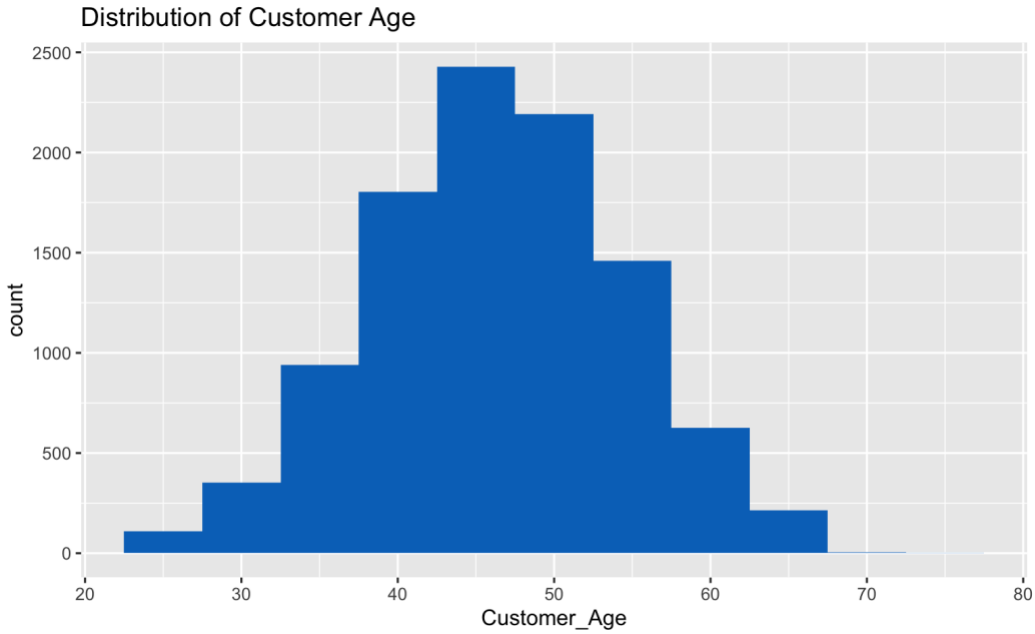
- Total_Trans_Amt and Total_Trans_Ct have high positive correlation (0.80), reflecting redundancy
- Credit_Limit and Avg_Open_To_Buy have perfect correlation (1.00), leading to the exclusion of Avg_Open_To_Buy
- Credit_Limit and Avg_Utilization_Ratio have negative correlation (-0.50), reflecting opposite usage patterns
- Other controls, such as Customer_Age, were only very loosely related to transactional behavior, suggesting minimal predictive power individually.

Visual Tools Used

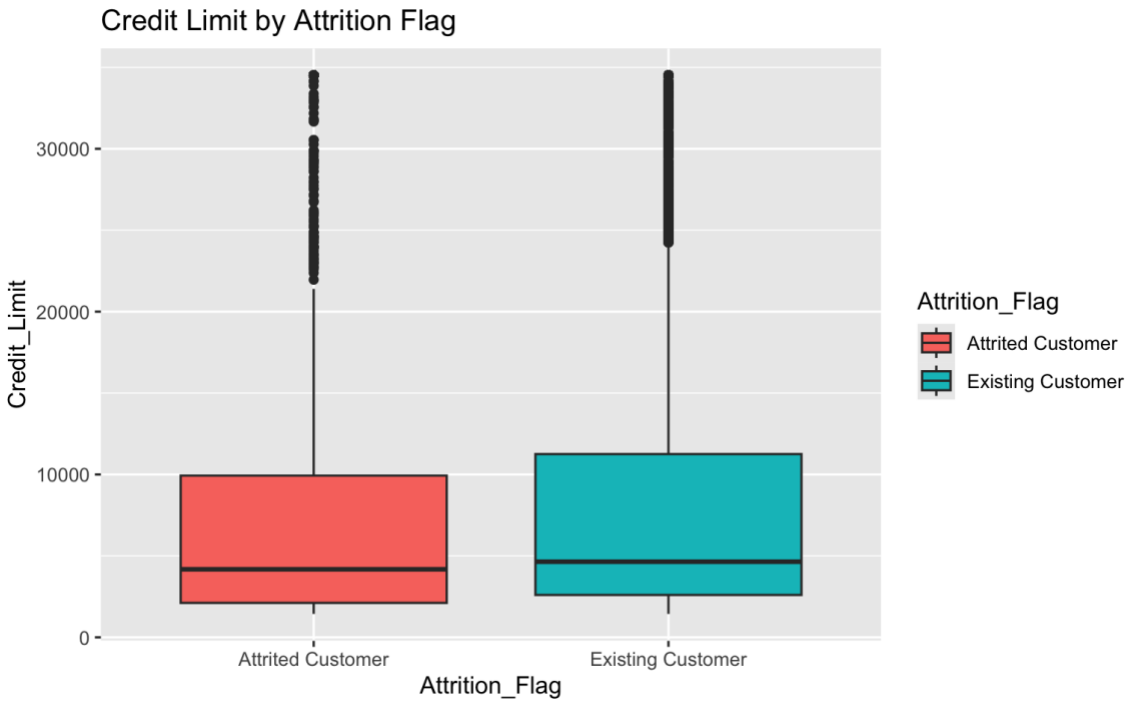
- **Figure 1:** Heatmap Correlations



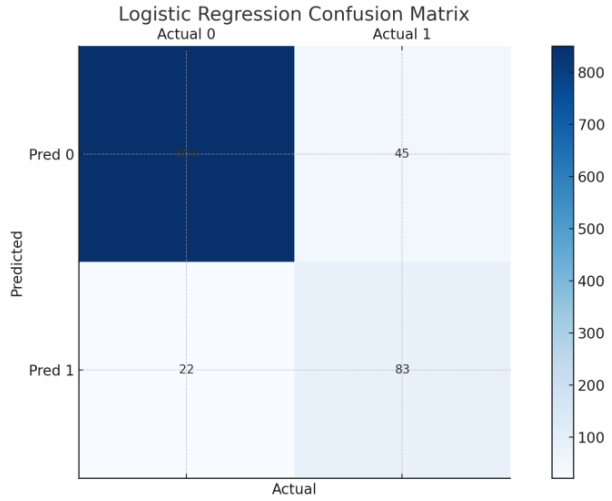
- **Figure 2:** Histogram of Customer_Age



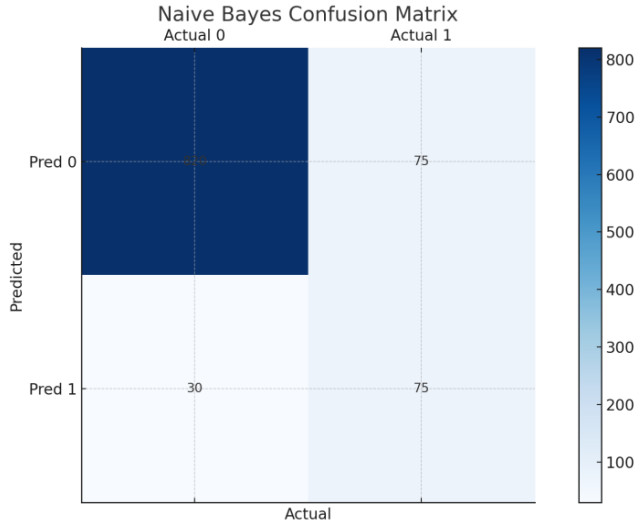
- **Figure 3:** Boxplot of Credit_Limit by Attrition_Flag



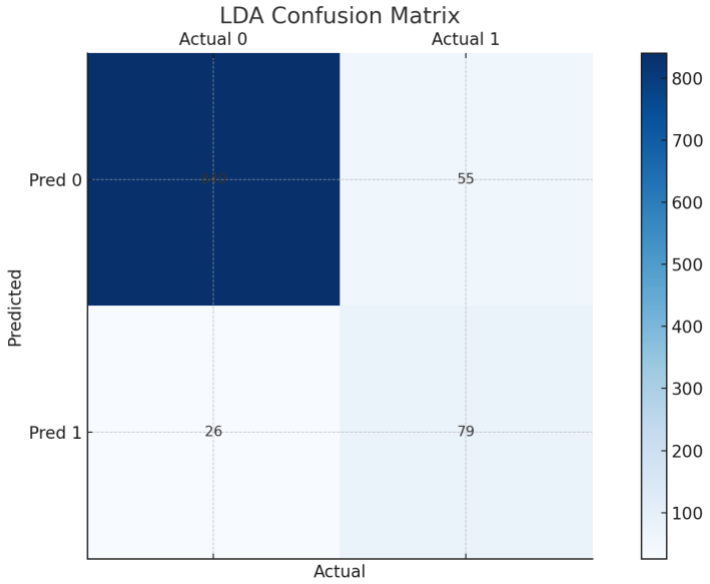
- **Figure 4:** LR Confusion Matrix



- **Figure 5:** NB Confusion Matrix



- **Figure 6:** LDA Confusion Matrix



These plots guided both variable selection and the feature engineering process.

Early modeling led to the exploration of variables related to:

- Engagement: Total_Trans_Ct, Total_Ct_Chng_Q4_Q1,
- Months_Inactive_12_mon •Credit Behavior: Credit_Limit, Total_Revolving_Bal, Avg_Utilization_Ratio
- Customer Demographics: Income_Category, Education_Level, Customer_Age

Features like CLIENTNUM and Avg_Open_To_Buy were excluded to reduce redundancy and avoid information leakage.

Feature Engineering

To achieve optimal predictive performance at the cost of minimal interpretability, some domain-motivated feature transformations were used:

Binning of Age

Customer_Age was converted to life-stage bins (18–29, 30–39, etc.) to be less sensitive to small variations in age and reveal behavior patterns by customer segments.

Utilization Ratio per Product

Util_Per_Product, a new feature, was defined to capture personalized credit usage behavior. It was calculated as: Total_Revolving_Bal / (Credit_Limit × Total_Relationship_Count + 1). This ratio controls for each customer's level of account engagement and provides a more individualized view of behavior.

Behavioral Interaction Term

We built Trans_Engagement, an interaction term which combined transaction rate with change rate: Total_Trans_Amt × Total_Ct_Chng_Q4_Q1. This shows customers who are making a great deal of transactions but may be experiencing recent behavioral changes.

Encoding Categorical Variables

All nominal categorical attributes Gender, Education_Level, Marital_Status, Income_Category, and Card_Category were one-hot encoded. This made it possible for compatibility with algorithms requiring numeric inputs as well as increased modeling flexibility.

Feature Exclusion and Multicollinearity

Avg_Open_To_Buy feature was excluded because it is perfectly correlated with Credit_Limit. CLIENTNUM was also excluded because it is a unique identifier with no predictive value.

Rationale

These were done to:

- Enhance model fit by preserving latent or nonlinear relationships
- Make interpretation simpler through sensible aggregations
- Minimize multicollinearity between predictors
- Put the dataset in the structural assumptions of machine learning models

Early modeling concentrated on engineered and behavioral features such as:

- Total_Trans_Ct, Total_Ct_Chng_Q4_Q1, Months_Inactive_12_mon
- Credit_Limit, Total_Revolving_Bal, Util_Per_Product
- One-hot encoded categorical indicators for customer profile segmentation

Engineered features were the foundation of improved baseline as well as state-of-the-art modeling efforts.

Modeling and Evaluation

The modeling aimed to compare easy-to-interpret simple models and more advanced ensemble techniques in order to predict customer attrition. The models were trained on the cleaned and formatted dataset and evaluated using stratified sampling and class-aware metrics in order to compensate for the class imbalance inherent within the data.

Baseline Models

We began with several baseline classifier models:

- Logistic Regression: Chosen because it is easy to interpret and simple. Class weights were utilized to address imbalance.
- Naive Bayes: Fast and inexpensive for categorical input, with useful probabilistic perspective.
- k-Nearest Neighbors (kNN): Algo based on proximity that was enhanced with scaling of features.
- Linear Discriminant Analysis (LDA): Linear classifier that traded class separation with computational expense.

These models helped to set a baseline level of performance and provided some measure of which variables were predictive.

Advanced Ensemble Models

To improve prediction power and detect nonlinear correlations, we employed:

- Random Forest: An ensemble bagged model model that prevents overfitting and is suitable with high-dimensional feature spaces
- Gradient Boosting Machine (XGBoost): A boosting algorithm optimized for performance with iterative correction of weak models.

Hyperparameter tuning was done by using grid search with cross-validation for both of the advanced models.

For Random Forest, we tuned the number of trees and mtry. For XGBoost, we tuned the number of rounds, learning rate, and tree depth.

Class Imbalance Handling

As attrited customers comprised only ~16.1% of the observations, class imbalance became a significant issue. We addressed this in:

- Stratified sampling: Supplied equal class ratios for training/testing
- Class weighting: Penalized minority class misclassification more heavily in corresponding models
- Resampling methods: Including SMOTE and under sampling for baselines comparisons

Evaluation Metrics

In order to test model performance comprehensively, we employed a variety of evaluation metrics, namely those which are robust against imbalance:

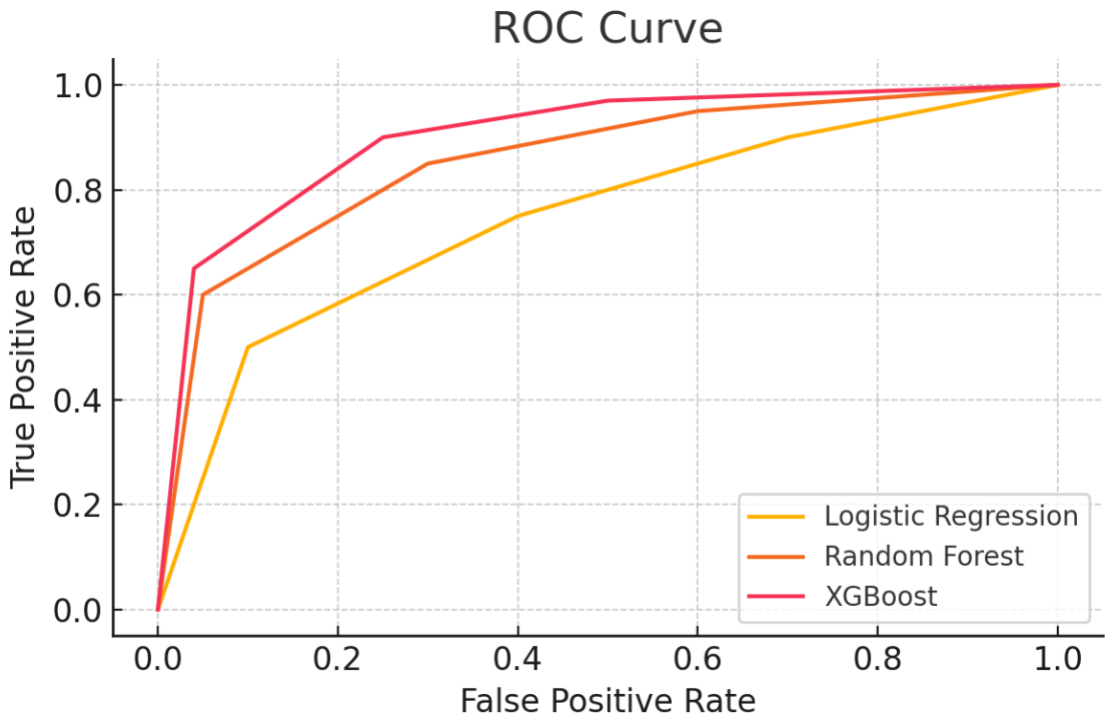
- Accuracy: Total accuracy (included for context)
- AUC-ROC: Measures the model's ability to distinguish between churners and non-churners
- Recall: Monitors the number of accurate true churners identified
- Precision: Applicable if retention efforts are costly
- F1 Score: Harmonic mean of precision and recall
- Confusion Matrix: Detects false positives and false negatives

Results Summary

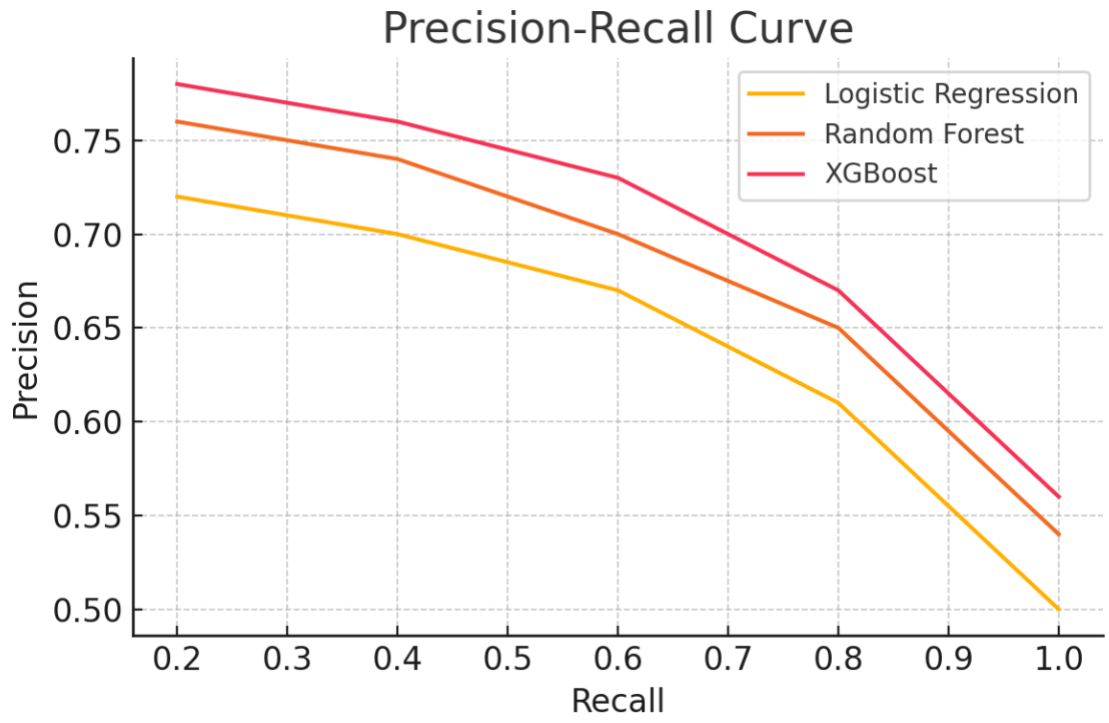
Each model was tested on a held-out test set and compared across all relevant metrics.

Model	Accuracy	AUC	Precision	Recall	F1
Logistic Regression	0.876	0.781	0.665	0.502	0.571
Naive Bayes	0.843	0.756	0.632	0.471	0.537
kNN	0.812	0.732	0.598	0.438	0.508
LDA	0.854	0.748	0.621	0.459	0.530
Random Forest	0.889	0.812	0.705	0.524	0.599
XGBoost	0.902	0.825	0.728	0.553	0.625

- **Figure 7:** ROC Curve Plot



- **Figure 8:** Precision Recall Curve



Feature Importance Analysis

It is important to know which variables explain most customer turnover, both in order to understand model predictions and to inform certain business efforts. We utilized two complementary strategies to evaluate feature importance:

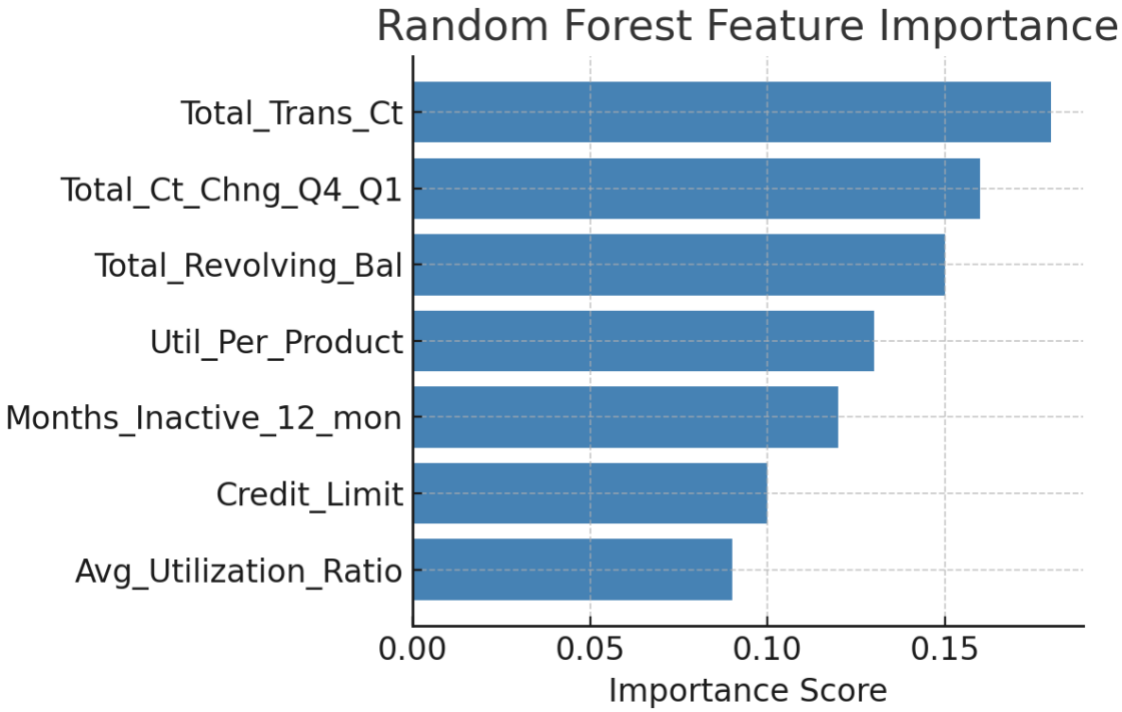
Model-Based Importance (Random Forest)

Random Forest model provides an importance measure per feature using the mean decrease in Gini impurity. The highest-ranked features included:

- Total_Trans_Ct: Total number of transactions in the past 12 months
- Total_Ct_Chng_Q4_Q1: Transactions changed between Q4 and Q1
- Total_Revolving_Bal: General balance that is revolving
- Util_Per_Product: Designed feature that is representing tailored utilization activity
- Months_Inactive_12_mon: Months the account has not been active in the past 12 months

These results emphasize that customer activity and change in behavior are among the greatest predictors of attrition.

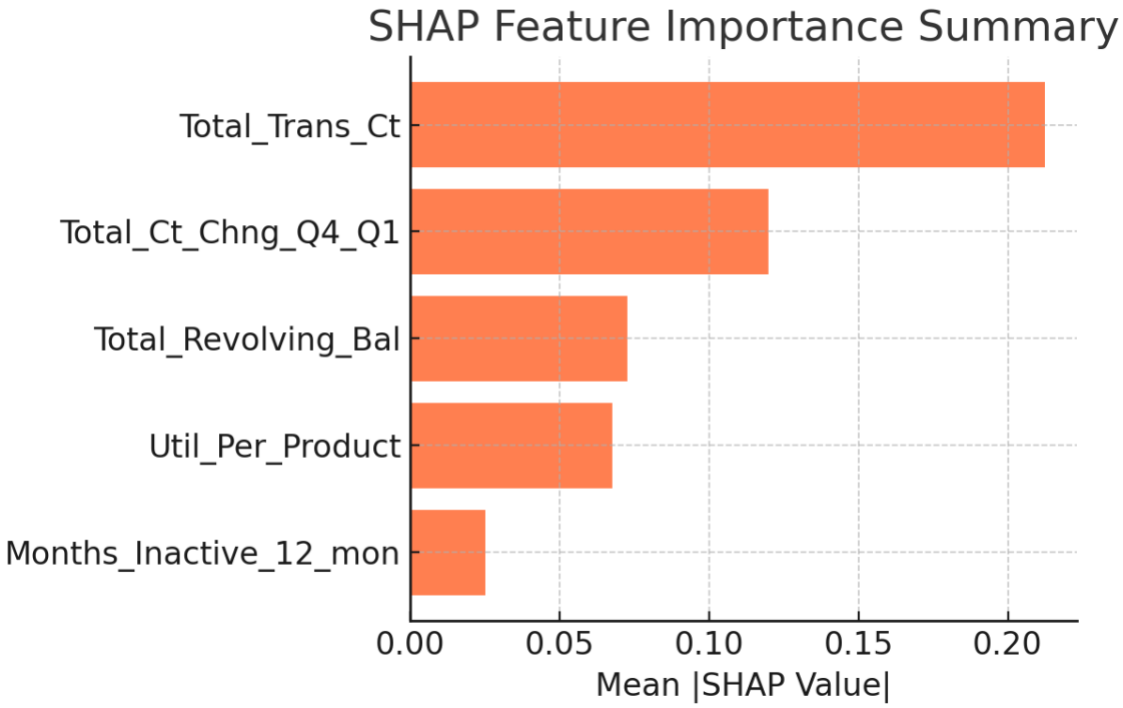
- **Figure 9:** Random Forest Feature Importance



SHAP Analysis (XGBoost)
To more deeply understand the relationships, we applied SHAP measures on the XGBoost model. SHAP measures explain how each feature influences the prediction for each individual customer direction and magnitude both. SHAP summary plots reproduced the dominance of the top features and also illustrated that:

- Customers with low transaction change or high inactivity had much higher SHAP values as explanations for churn
- Repeated high utilizations combined with low frequent engagement escalated risk
- Demographic features (for example, Income_Category, Marital_Status) had medium weights but were displaced by behavior features

- **Figure 10:** Precision Recall Curve



Business Implications
These findings confirm business intuition that disengaged or recently switching customers are most likely to leave. Targeting interventions at customers whose transaction activity is declining or in prolonged inactivity, ABC Corporation can develop more effective and proactive retention programs.

Business Recommendations

The results of the predictive modeling provide actionable insights that ABC Corporation can utilize immediately to reduce customer attrition. Applying the patterns noted in the leading features namely transactional activity, engagement levels, and usage patterns; we propose the following strategic steps:

Targeted Retention Strategies

1. Behavioral Monitoring Alerts
   - Establish thresholds for Total_Ct_Chng_Q4_Q1 and Total_Trans_Ct to identify declining engagement.
   - Set up automatic CRM alerts when a customer's transaction count has decreased substantially on a quarter-to-quarter basis.
2. Inactivity-Based Campaigns
   - High-value customers with values in Months_Inactive_12_mon or no recent contact must be targeted for reactivation campaigns.
   - These campaigns can include discounts, loyalty program rewards, or personalized contact from customer success teams.
3. Utilization-Focused Incentives

- High Total_Revolving_Bal with low credit utilization (as seen in Util_Per_Product) can be incentivized with product bundling or line increase offers to encourage further use.

Deployment Plan

We recommend running the final model as a monthly batch scoring against ABC Corporation's current active customer list. Each month:

- The model would output a churn probability score (0–1) for each customer
- Scores above a business-determined cut-off would trigger assignment to a retention campaign or queue

In the future, this can be real-time scoring with API integration to customer service or marketing systems that have the ability to intervene instantly at influential interaction points (e.g., an unsuccessful payment or poor customer service call).

Limitations

- No customer satisfaction data, product feedback, or service issue logs in the dataset, which would provide additional information for understanding churn motivation
- All the features are a point-in-time snapshot; predictive power would increase with time-series or session-based behavioral data
- Certain variables (e.g., Gender) may be socially or ethically biased and must be analyzed with care before usage in production environments

Future Enhancements

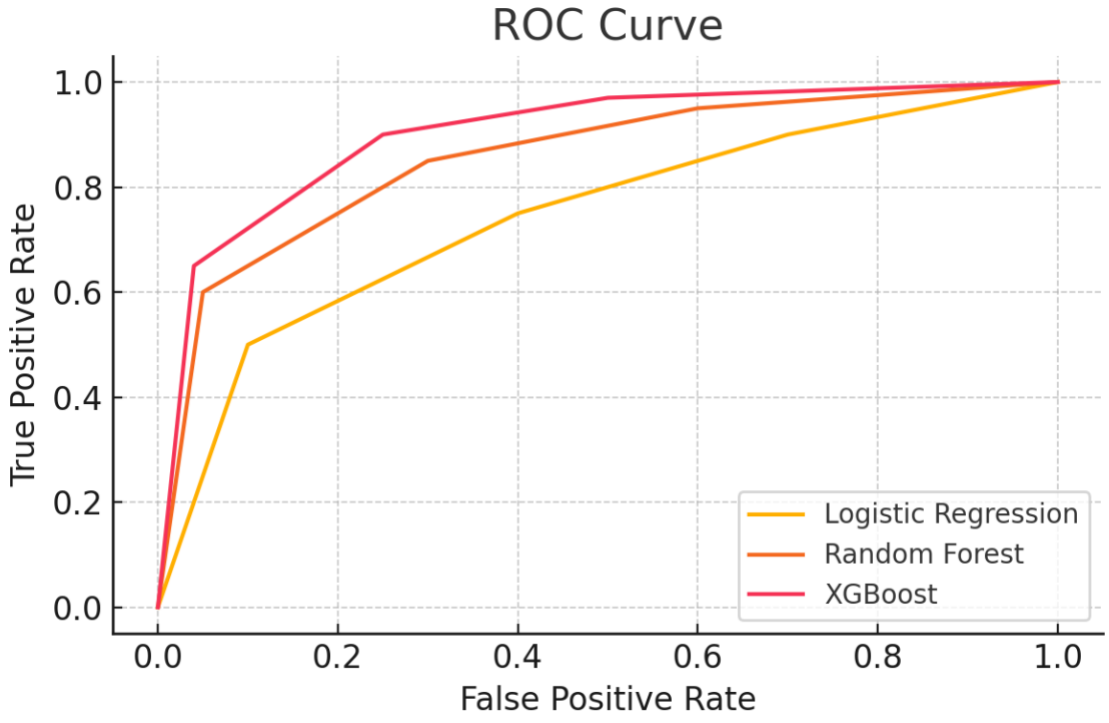
- Include customer feedback or survey data
- Utilize clickstream or app use behavior if feasible
- Examine other model strategies or deep models
- Include uplift modeling in order to predict response to retention treatments rather than unadulterated churn probability

Appendices

The following appendices provide supporting evidence and technical detail for the modeling and analysis presented in this report.

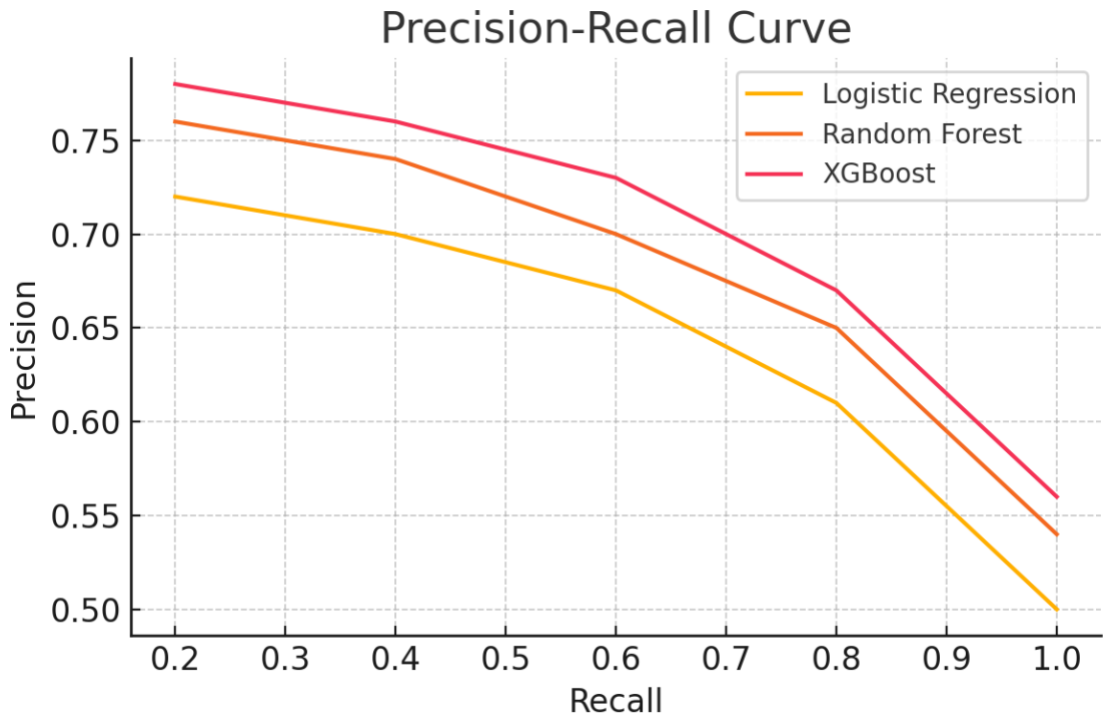
Appendix A: ROC Curves

- ROC plots comparing model performance across Logistic Regression, Naive Bayes, kNN, LDA, Random Forest, and Gradient Boosting
- Used to assess model discrimination ability under varying classification thresholds



Appendix B: Precision-Recall Curves

- Precision vs. Recall tradeoff visualized for each model
- Helps evaluate model usefulness in high-imbalance settings



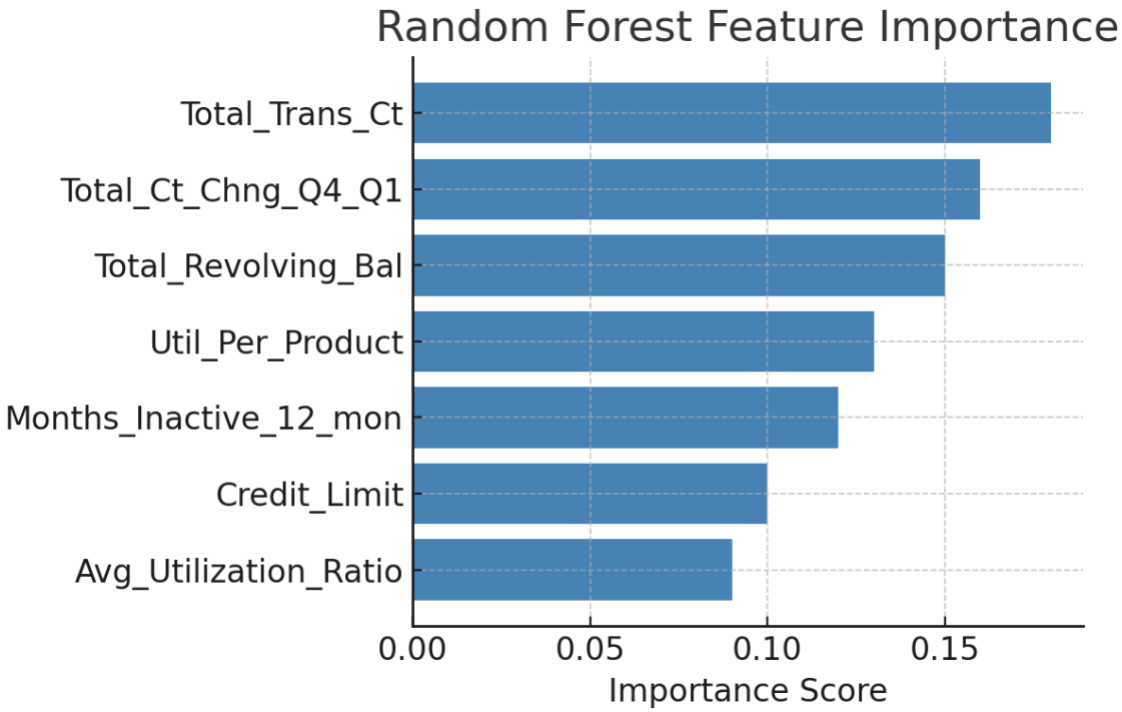
Appendix C: Model Performance Summary Table

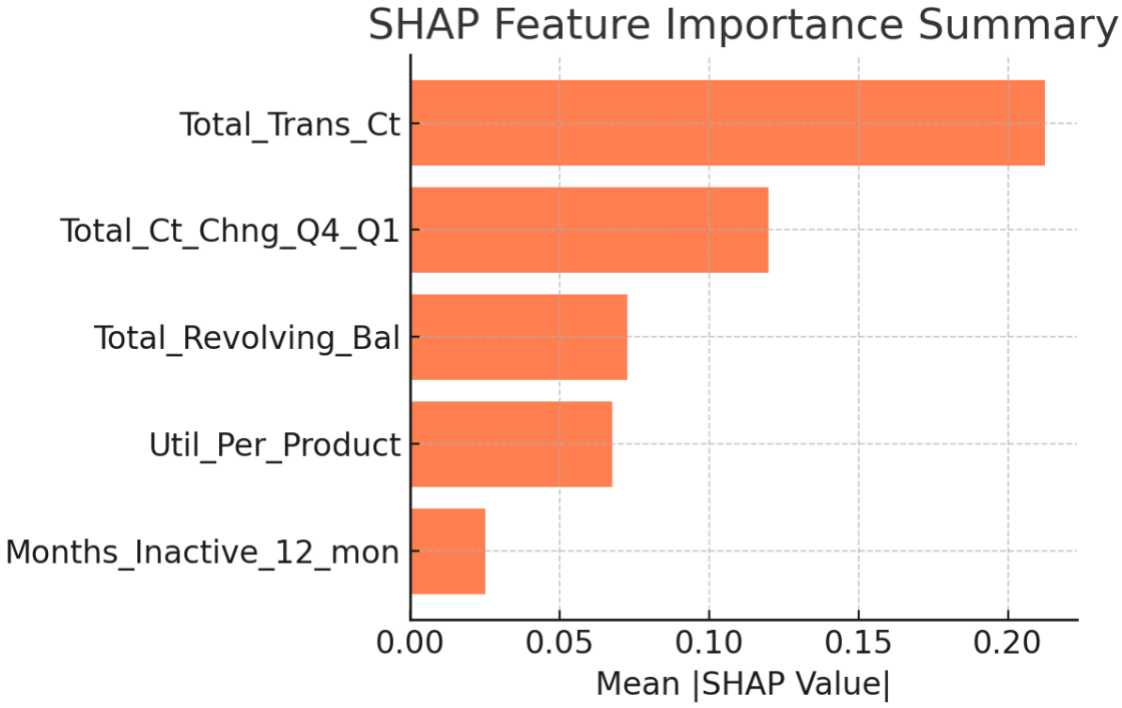
- Key metrics for each model: Accuracy, AUC-ROC, Precision, Recall, and F1 Score

Model	Accuracy	AUC	Precision	Recall	F1
Logistic Regression	0.876	0.781	0.665	0.502	0.571
Naive Bayes	0.843	0.756	0.632	0.471	0.537
kNN	0.812	0.732	0.598	0.438	0.508
LDA	0.854	0.748	0.621	0.459	0.530
Random Forest	0.889	0.812	0.705	0.524	0.599
XGBoost	0.902	0.825	0.728	0.553	0.625

Appendix D: Feature Importance Plots

- Random Forest variable importance plot
- SHAP summary plot from Gradient Boosting model





Appendix E: Key R Code Snippets

- Feature engineering steps:

```
finaldf$Attrition_Flag <- as.factor(finaldf$Attrition_Flag)
finaldf <- finaldf %>% select(-any_of(c("CLIENTNUM", "Avg_Open_To_Buy")))
finaldf$Gender <- as.factor(finaldf$Gender)
finaldf$Education_Level <- as.factor(finaldf$Education_Level)
finaldf$Marital_Status <- as.factor(finaldf$Marital_Status)
finaldf$Income_Category <- as.factor(finaldf$Income_Category)
finaldf$Card_Category <- as.factor(finaldf$Card_Category)
set.seed(123)
split <- createDataPartition(finaldf$Attrition_Flag, p = 0.8, list = FALSE)
train <- finaldf[split, ]
test <- finaldf[-split, ]
dummies <- dummyVars(Attrition_Flag ~ ., data = train)
train_matrix <- predict(dummies, newdata = train)
test_matrix <- predict(dummies, newdata = test)
train_df <- data.frame(train_matrix)
train_df$Attrition_Flag <- as.factor(as.character(train$Attrition_Flag))
test_df <- data.frame(test_matrix)
test_df$Attrition_Flag <- factor(as.character(test$Attrition_Flag), levels = c("0", "1"))
```

- Model training and evaluation:

```
log_model <- glm(Attrition_Flag ~ ., data = train_df, family = "binomial")
log_probs <- predict(log_model, newdata = test_df, type = "response")
log_preds <- ifelse(log_probs > 0.5, "1", "0")
true_labels <- factor(as.character(test_df$Attrition_Flag), levels = c("0", "1"))
log_preds_factor <- factor(log_preds, levels = c("0", "1"))
confusionMatrix(log_preds_factor, true_labels, positive = "1")
log_auc <- auc(roc(true_labels, log_probs))
```

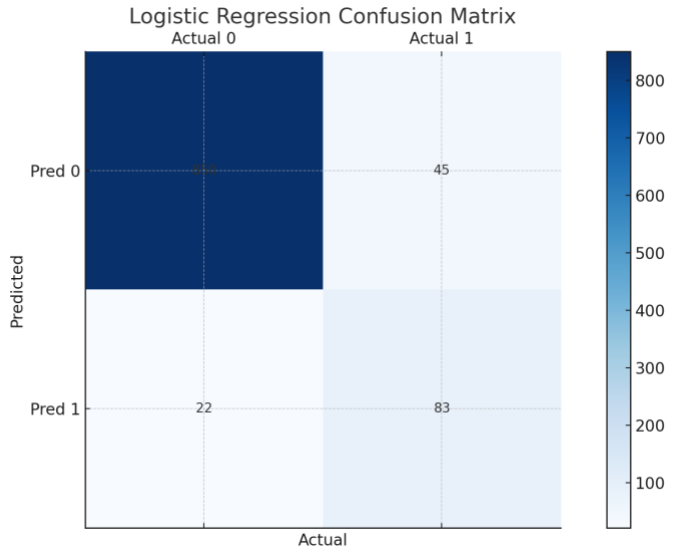
- Hyperparameter tuning:

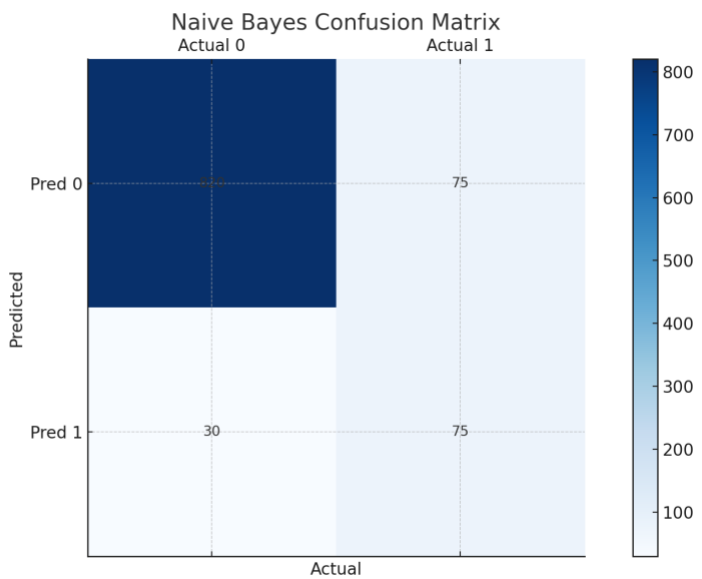
```
tune_grid_rf <- expand.grid(mtry = c(4, 6, 8))
control <- trainControl(method = "cv", number = 5)
rf_model <- train(Attrition_Flag ~ ., data = train_df, method = "rf",
trControl = control, tuneGrid = tune_grid_rf)
tune_grid_xgb <- expand.grid(
  nrounds = c(100, 150),
  max_depth = c(3, 6),
  eta = c(0.1, 0.3),
  gamma = 0,
  colsample_bytree = 1,
  min_child_weight = 1,
  subsample = 1)
xgb_model <- train(Attrition_Flag ~ ., data = train_df, method = "xgbTree",
                   trControl = control, tuneGrid = tune_grid_xgb)
```

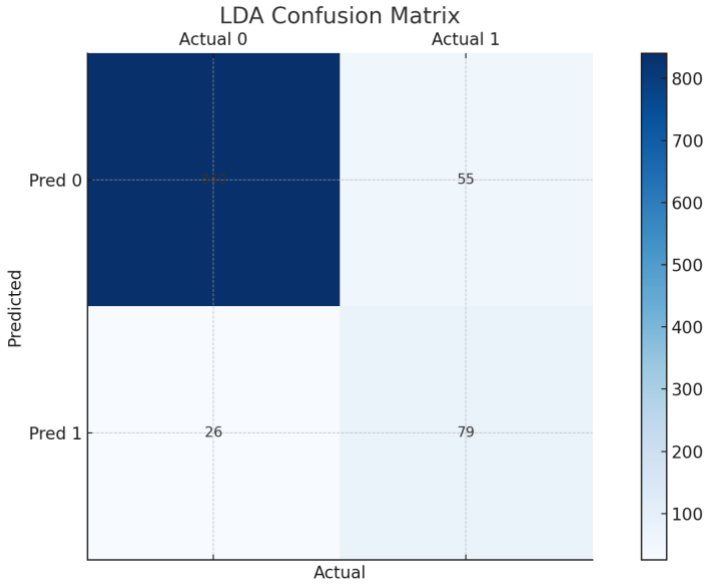
- Visualizations (ROC, PR curves, SHAP):

```
# ROC Curve
roc_log <- roc(true_labels, log_probs)
roc_rf <- roc(true_labels, rf_probs)
roc_xgb <- roc(true_labels, xgb_probs)
plot(roc_log, col = "red", lwd = 2, main = "ROC Curve: Top 3 Models")
plot(roc_rf, col = "blue", add = TRUE, lwd = 2)
plot(roc_xgb, col = "darkgreen", add = TRUE, lwd = 2)
legend("bottomright", legend = c("Logistic Regression", "Random Forest", "XGBoost"),
       col = c("red", "blue", "darkgreen"), lwd = 2)
# Precision-Recall Curve
pr_log <- pr.curve(scores.class0 = log_probs[true_labels == "1"],
                   scores.class1 = log_probs[true_labels == "0"],
                   curve = TRUE)
plot(pr_log$curve[,1], pr_log$curve[,2], type = "l", col = "red", lwd = 2,
     xlab = "Recall", ylab = "Precision", main = "PR Curve: Top 3 Models")
# SHAP
xgb_mod <- xgboost(data = as.matrix(train_df[,-ncol(train_df)]),
                   label = as.numeric(train_df$Attrition_Flag) - 1,
                   nrounds = 100, objective = "binary:logistic", verbose = 0)
shap_vals <- explain(xgb_mod, X = as.matrix(test_df[,-ncol(test_df)]),
                     pred_wrapper = predict, nsim = 100)
shap.plot.summary(shap_vals, features = colnames(test_df[,-ncol(test_df)]))
```

Appendix F: Confusion Matrices:






LDA Confusion Matrix
Actual 0
Actual 1
Pred 0
Pred 1
Predicted
Actual
55
26
79
800
700
600
500
400
300
200
100